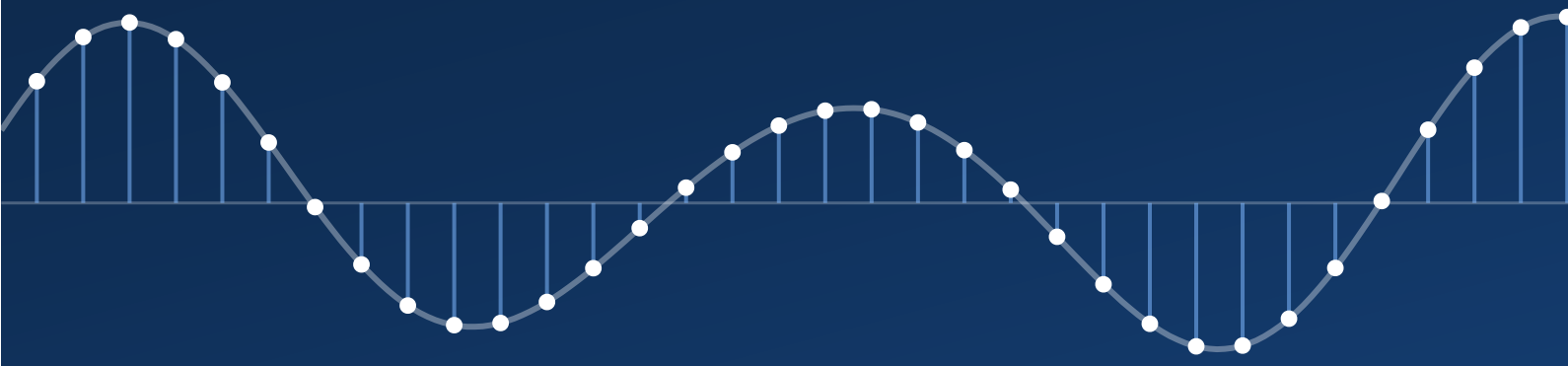
DIGITAL SIGNAL PROCESSING

CHAPTER 01

حل Sheet 1

حل نموذجي لكل أسئلة Sheet 1 (1.1 لحد 1.8)، بجُمل الدكتور من المحاضرة وبطريقته في الحل، زي ما يتكتب في ورقة الإجابة.

Sec. 1.1 – 1.4 | Sec. 1.5 | Sec. 1.6 | Example 1.3



8	5	3
أسئلة	نظري	مسائل

المحتوى

01 **الجزء النظري**
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إزاي تستخدم الملف

النظري بجُمل الدكتور من المحاضرة، مترتب عشان يبان إنك فاهم. والمسائل خطوة بخطوة بنفس طريقة الدكتور في أمثلة المحاضرة. السطر البرتقالي بيقولك الحل جاي من أنهي جزء في المحاضرة.

SECTION 01

الجزء النظري

أسئلة 1.1 لحد 1.5

1. Define: Signal, System, Signal processing (state one example for each).

2. Classify the different types of signal processing.

3. Classify the different signal types.

4. Explain the basic three operations performed in A/D converters.

5. State and explain the sampling theorem.

1.1 Define: Signal, System, Signal processing (state one example for each).

1.1 Define the following items: Signal, system, signal processing. (State one example for each).

SOLUTION

- **Signal:** is defined as any physical quantity that varies with time, space, or any other independent variable(s). Mathematically, we describe a signal as a function of one or more independent variables.
  Examples: $s1(t) = 5t$ (varies linearly with t), $s2(t) = 20t^2$ (varies quadratically with t), and $s(x, y) = 3x + 2xy + 10y^2$ (two independent variables, e.g. the spatial coordinates in a plane). A speech signal cannot be described by such an expression; a segment of it is represented as a sum of several sinusoids of different amplitudes, frequencies and phases.
- **System:** is defined as a physical device that performs an operation on a signal.
  Example: a filter used to reduce the noise and interference corrupting a desired information. If the operation is linear, the system is called linear; if it is nonlinear, the system is nonlinear.
- **Signal processing:** when we pass a signal through a system, as in filtering, we say that we have processed the signal.
  Example: filtering the noise and interference from the desired signal. Digital processing may be performed by digital hardware (logic circuits), by software, or by a combination of both.

Sec. 1.1 Lec 2 المرجع

1.2 Classify the different types of signal processing.

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SOLUTION

There are two types of signal processing systems (Sec. 1.2):

1. **Analog signal processor (Fig. 1.1):** the analog input signal is processed directly by an analog signal processor to give an analog output signal.
2. **Digital signal processor (Fig. 1.2):** provides an alternative method for processing the analog signal. The **A/D converter** converts the analog input into a digital signal (a sequence of numbers), the **digital signal processor** performs the required operations on it, and the **D/A converter** converts the result back into an analog output signal.





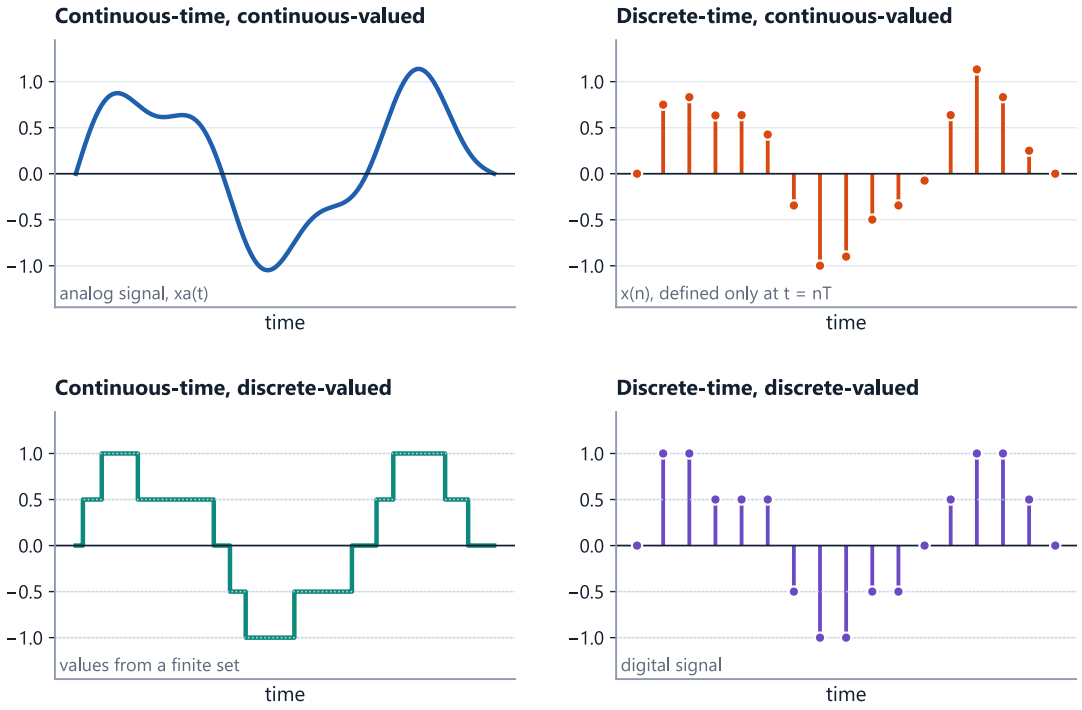
Fig. 1.1, 1.2 Sec. 1.2 Lec 2 المرجع

1.3 Classify the different signal types.

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SOLUTION

1. **Continuous-time vs. discrete-time:** continuous-time (analog) signals are defined for every value of time and take on values in the continuous interval (a, b). Discrete-time signals are defined only at certain specific values of time.
2. **Continuous-valued vs. discrete-valued:** a continuous-valued signal takes on all possible values on a finite or an infinite range. A discrete-valued signal takes on values from a finite set of possible values.
3. **Deterministic vs. random:** a deterministic signal can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule. Otherwise, the signal is random.
4. **Multichannel vs. multidimensional:** a multichannel signal is generated by multiple sources or sensors and represented in vector form, $S3(t) = [s1(t)\ s2(t)\ s3(t)]^T$. A signal is M-dimensional if its value is a function of M independent variables.



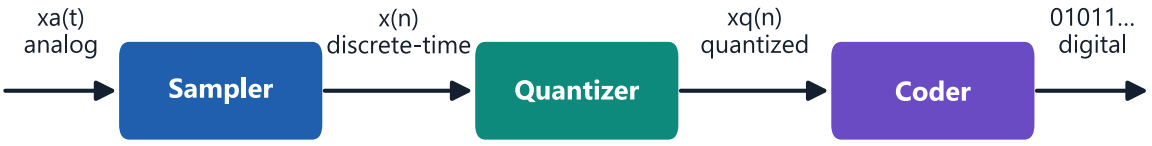
Sec. 1.4 Lec 2 المرجع

1.4 Explain the basic three operations performed in A/D converters.

1.4 Explain the basic three operations performed in A/D converters.

SOLUTION

To process analog signals by digital means, it is first necessary to convert them into a sequence of numbers having finite precision. This is A/D conversion, a three-step process (Fig. 1.9):



1. **Sampling:** the conversion of a continuous-time signal into a discrete-time signal obtained by taking "samples" of the continuous-time signal at discrete-time instants. If xa(t) is the input to the sampler, the output is **x(n) = xa(nT)**, where T is the sampling interval.
2. **Quantization:** the conversion of a discrete-time continuous-valued signal into a discrete-time, discrete-valued (digital) signal. The value of each sample is selected from a finite set of possible values. The difference $x_q(n) - x(n)$ is the quantization error.
3. **Coding:** each discrete value $x_q(n)$ is represented by a **b-bit binary sequence**.

Sampling does not result in a loss of information if the signal bandwidth is finite (**invertible**), while quantization is a **noninvertible** process that results in signal distortion.

Fig. 1.9 Sec. 1.6 Lec 4 المرجع

1.5 State and explain the sampling theorem.

1.5 State and explain the sampling theorem.

SOLUTION

- To select Fs we must have information about the frequency content of the signal, e.g. f_{max} = 3000 Hz for speech and 5 MHz for television signals. The signal may be passed through a filter that severely attenuates components above f_{max}.
- To avoid aliasing, Fs is selected so that **$Fs > 2f_{max}$** (1.47). Then every component $|f_i| < f_{max}$ is mapped into $-1/2 \le F_i = f_i/Fs \le 1/2$ (1.48).

Sampling theorem: if the highest frequency contained in an analog signal xa(t) is $f_{max} = B$ and the signal is sampled at a rate $Fs > 2f_{max} = 2B$, then xa(t) can be exactly recovered from its sample values using the interpolation function:

$$g(t) = \sin(2\pi Bt) / (2\pi Bt) \quad (1.50) \qquad xa(t) = \Sigma\, xa(n/Fs)\, g(t - n/Fs) \quad (1.51)$$

- The sampling rate $\mathbf{f_N = 2B = 2f_{max}}$ is called the **Nyquist rate**.
- Because of the complexity and the infinite number of samples required, these reconstruction formulas are primarily of theoretical interest.

Sec. 1.6.2 Lec 4 **المرجع**

SECTION 02

المسائل

أسئلة 1.6 لحد 1.8

1. Which sinusoids are periodic? Compute their fundamental period.

2. xa(t) = 3 sin(100πt): sketch, sample at Fs = 300, period, peak value

3. A/D with T1 = 5 ms then D/A with T2 = 1 ms: find ya(t)

1.6 Which sinusoids are periodic? Compute their fundamental period.

1.6 Determine which of the following sinusoid signals are periodic, and then compute their fundamental period.
(a) $x_a(t) = 3\cos(5t + \pi/6)$ (b) $x_a(t) = 3\cos(100\pi t)$ (c) $x(n) = \cos[\pi(30n/105)]$ (d) $x(n) = 2\exp[j(n/6 - \pi)]$

SOLUTION

(a)

$x_a(t) = 3\cos(5t + \pi/6)$ is a continuous-time sinusoid $x_a(t) = A\cos(2\pi ft + \theta)$. For every fixed value of f, $x_a(t)$ is periodic with $T_p = 1/f$.

$$2\pi f = 5 \rightarrow f = 5/2\pi \text{ Hz} \rightarrow \mathbf{T_p = 1/f = 2\pi/5 \text{ sec}}$$
$$x_a(t + T_p) = 3\cos(5(t + 2\pi/5) + \pi/6) = 3\cos(5t + 2\pi + \pi/6) = x_a(t)$$

∴ Periodic, fundamental period $T_p = 2\pi/5$ sec.

(b)

$x_a(t) = 3\cos(100\pi t)$ is a continuous-time sinusoid $x_a(t) = A\cos(2\pi ft + \theta)$. For every fixed value of f, $x_a(t)$ is periodic with $T_p = 1/f$.

$$2\pi f = 100\pi \rightarrow f = 50 \text{ Hz} \rightarrow \mathbf{T_p = 1/f = 20 \text{ msec}}$$
$$x_a(t + T_p) = 3\cos(100\pi(t + 20 \text{ ms})) = 3\cos(100\pi t + 2\pi) = x_a(t)$$

∴ Periodic, fundamental period $T_p = 20$ msec.

(c)

$x(n) = \cos[\pi(30n/105)]$ is a discrete-time sinusoid $x(n) = A\cos(2\pi Fn + \theta)$: it is periodic only if F is rational ($F = k/N$).

$$2\pi F = 30\pi/105 \rightarrow F = 1/7 = k/N \text{ (k and N relatively prime)} \rightarrow \text{rational}$$
$$x(n + 7) = \cos(30\pi(n + 7)/105) = \cos(30\pi n/105 + 2\pi) = x(n)$$

∴ Periodic, fundamental period **N = 7 samples**.

(d) $x(n) = 2\exp[j(n/6 - \pi)] = 2e^{-j\pi}e^{j2\pi Fn}$: a discrete-time complex exponential is periodic only if its relative frequency is rational.

$$2\pi F = 1/6 \rightarrow F = 1/(12\pi)$$

F cannot be expressed as a ratio of two integers → **not periodic**.

المرجع: Sec. 1.5.3.2 · Sec. 1.5.2 (Eq. 1.16) · Sec. 1.5.1 · Lec 3

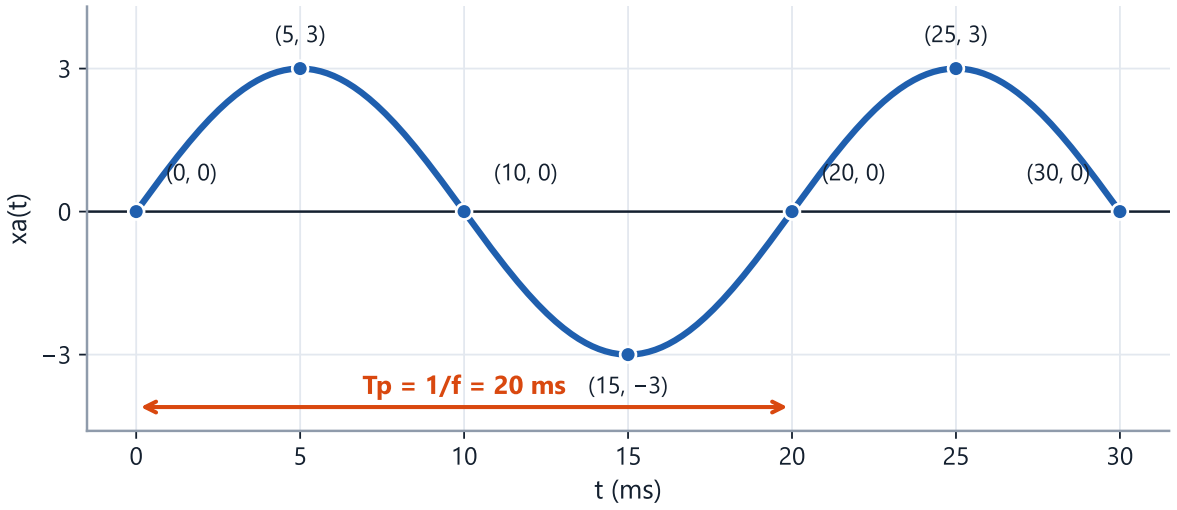
1.7 xa(t) = 3 sin(100πt): sketch, sample at Fs = 300, period, peak value

1.7 Consider the analog sinusoid signal xa(t) = 3 sin(100πt).
(a) Sketch the signal for [0, 30] ms.
(b) The signal is sampled with sampling rate Fs = 300 samples per second. Determine the frequency of the discrete-time signal and show that it is periodic.
(c) Compute the sample values in one period of x(n). Sketch x(n) on the same diagram with xa(t). What is the period of this discrete-time signal x(n)?
(d) Can you find a sampling rate Fs such that the signal x(n) reaches its peak value 3? What is the minimum Fs suitable for this task?

SOLUTION

(a) xa(t) = 3 sin(100πt) = A sin(2πft): A = 3, 2πf = 100π → f = 50 Hz, Tp = 1/f = 20 msec. The interval [0, 30] ms contains 1.5 periods.

1.7 (a) xa(t) = 3 sin(100πt) on [0, 30] ms (f = 50 Hz, 1.5 cycles)



(b)

If the signal is sampled at Fs = 300 Hz (t = nT = n/Fs), the discrete-time signal is:

$$x(n) = x_a(nT) = x_a(n/F_s) = 3\sin(100\pi\, n/300) = \mathbf{3\sin(\pi/3\, n)}$$

$$F = f/F_s = 50/300 = \mathbf{1/6} = k/N_P \;\rightarrow\; \text{rational} \rightarrow \text{periodic}, \quad \mathbf{N_P = 6 \text{ samples}}$$

$$x(n+6) = 3\sin(\pi(n+6)/3) = 3\sin(\pi n/3 + 2\pi) = x(n)$$

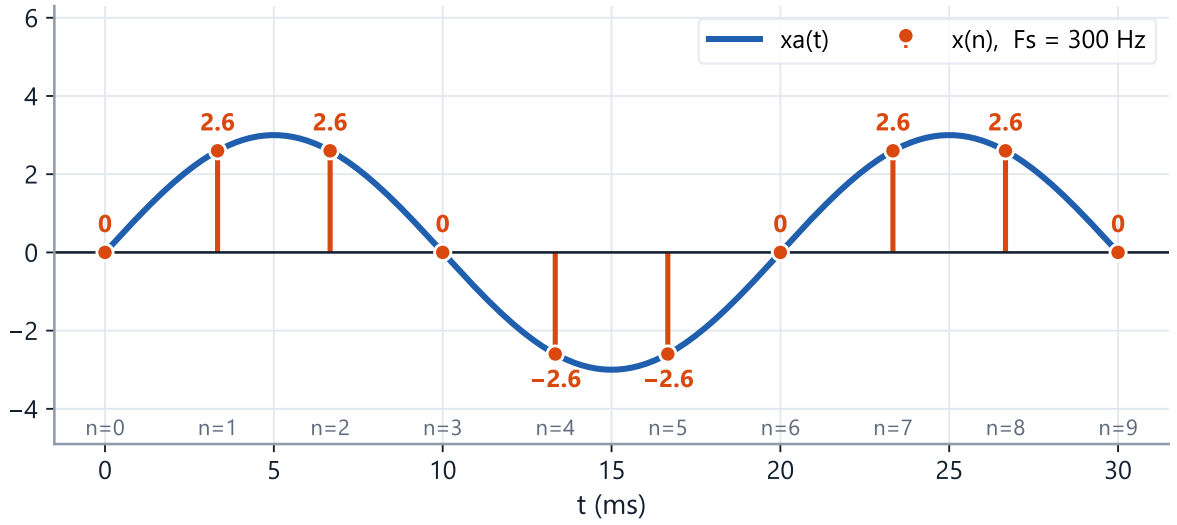
(c)

$$x(0) = 3\sin(0) = \mathbf{0}$$
$$x(1) = 3\sin(\pi/3) = \mathbf{2.598}$$
$$x(2) = 3\sin(2\pi/3) = \mathbf{2.598}$$

$$x(3) = 3\sin(\pi) = \mathbf{0}$$
$$x(4) = 3\sin(4\pi/3) = \mathbf{-2.598}$$
$$x(5) = 3\sin(5\pi/3) = \mathbf{-2.598}$$

The period of x(n) is **N = 6 samples** = 6 × (1/300) s = 20 msec.

1.7 (c) Fs = 300 Hz → x(n) = 3 sin(πn/3), N = 6 samples = 20 ms

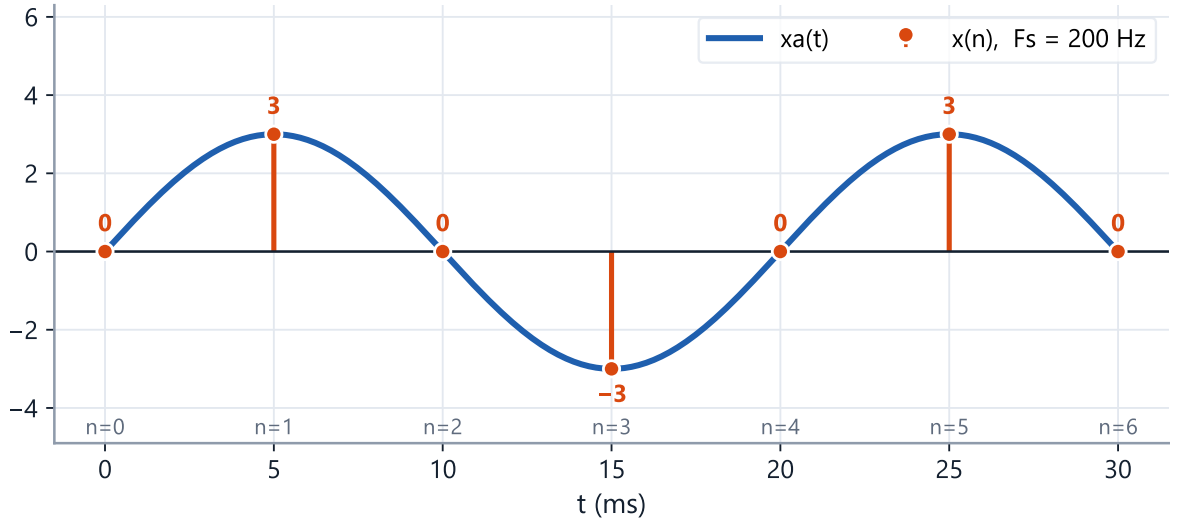


(d) x(n) = 3 sin(100πn/Fs) reaches its peak value 3 when:

$$\sin(100\pi n/Fs) = 1 \rightarrow 100\pi n/Fs = \pi/2 \rightarrow Fs = 200n$$

Yes. The minimum is at n = 1: **Fs = 200 samples/sec**. Then x(n) = 3 sin(πn/2) = 0, 3, 0, −3.

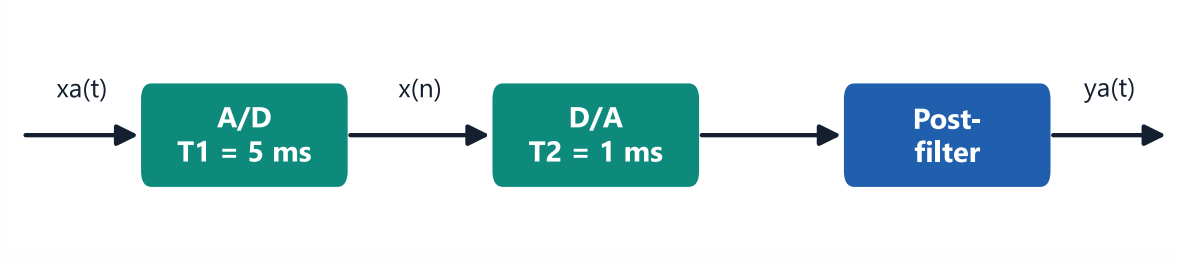
1.7 (d) Fs = 200 Hz → x(n) = 3 sin(πn/2) reaches the peak 3



Eq. 1.30, 1.33 Example 1.1 Lec 4 المرجع

1.8 A/D with T1 = 5 ms then D/A with T2 = 1 ms: find ya(t)

1.8 Consider a simple signal processing system as shown in the following figure. The sampling periods of the A/D and D/A converters are T1 = 5 ms and T2 = 1 ms respectively. Determine the output ya(t) if the input is given by: xa(t) = 3 cos(100πt) + 2 sin(250πt) (t in second).



SOLUTION

The frequencies existing in the analog signal are f1 = 50 Hz (2πf1 = 100π) and f2 = 125 Hz (2πf2 = 250π).

The A/D converter has T1 = 5 ms, so Fs = 1/T1 = 200 samples/s and the folding frequency is Fs/2 = 100 Hz, the maximum frequency that can be represented uniquely by the sampled signal. By making use of (1.30) we obtain:

$$\begin{aligned} x(n) &= x_a(nT1) = x_a(n/Fs) \\ &= 3\cos 2\pi(50/200)n + 2\sin 2\pi(125/200)n \\ &= 3\cos 2\pi(1/4)n + 2\sin 2\pi(5/8)n \\ &= 3\cos 2\pi(1/4)n + 2\sin 2\pi(1-3/8)n \\ &= 3\cos 2\pi(1/4)n + 2\sin 2\pi(-3/8)n \end{aligned}$$

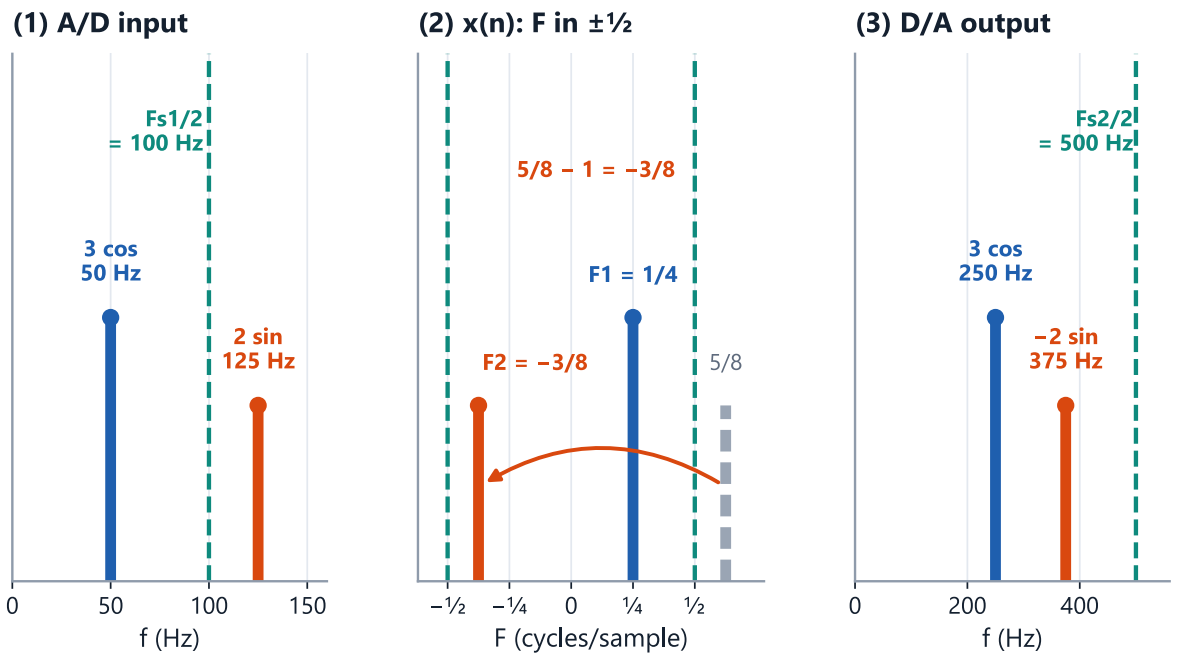
Finally, we obtain:

$$\mathbf{x(n) = 3\cos(\pi n/2) - 2\sin(3\pi n/4)}$$

The D/A converter has T2 = 1 ms (Fs = 1000 samples/s). Using t = nT2, i.e. n = t/T2 = 1000t:

$$y_a(t) = 3\cos(\pi(1000t)/2) - 2\sin(3\pi(1000t)/4) = \mathbf{3\cos(500\pi t) - 2\sin(750\pi t)}$$

which is different from xa(t): the distortion is caused by aliasing (Fs = 200 Hz < 2 × 125 Hz), and the frequencies are scaled by T1/T2 = 5.



Eq. 1.30 Example 1.3 Lec 4 المرجع